AGENTVERSE 2026 · BIOTECH TRACK · TEAM BACKPROP TO THE FUTURE

Which drug pairs should a lung cancer lab test next?

5,460

possible drug pairs from 105 NCI-ALMANAC drugs

Budget for 50.

In NCI-ALMANAC screens across 9 lung cancer cell lines, only 172 of 5,228 measured pairs (3.3%) show synergy. Over 96% are dead ends.

LungPair ranks drug combinations for NSCLC labs and deterministically refuses when evidence runs out.

THE PROBLEM

Three bottlenecks between the lab and the patient

1. Too many pairs, too little budget

Testing all 5,228 measured pairs across 9 lung cell lines requires ~47,000 experiments—costing roughly $16M at $340 per experiment ($3,060/pair).

Labs choose pairs using literature review and intuition, wasting >96% of robotic capacity on dead ends.

2. Single drugs stop working

Systemic toxicity limits monotherapy dosing, and acquired resistance inevitably emerges in metastatic non-small cell lung cancer (NSCLC).

Combination regimens overcome resistance mechanisms, but the search space explodes combinatorially.

3. Old drugs, new pairs

Repurposing approved compounds leverages known safety and pharmacokinetic profiles, dramatically accelerating the path to Phase II.

Rather than de novo small-molecule discovery, intelligent doublet triage creates rapid clinical translational hypotheses.

Repurposing pathway: Kulkarni et al., Russ J Bioorg Chem 2023. Core facility cost: $340/experiment (1 pair in 9 cell lines = $3,060).

WHO IT'S FOR

Built for lab researchers, not patients

Academic NSCLC Labs

Bench researchers testing drug combinations in vitro.

Lab Technicians & Translational Teams

Core teams programming high-throughput liquid-handling robots.

Screening Cores & CROs (B2B SaaS)

Institutional screening facilities recharging on hit-rate yield.

What LungPair gives them

- A ranked shortlist of high-probability candidate pairs
- Calibrated confidence tiers backed by held-out precision metrics
- Deterministic refusal when chemical evidence runs out (Tier 4)
- Mandatory human scientist sign-off before robotic assay setup
- Cryptographically auditable JSONL execution receipts

RESEARCH USE ONLY (RUO). In vitro cell-line screening prioritization. Bypasses hospital procurement drag—zero clinical diagnosis or treatment claims.

THE PIPELINE

From public screens to a shortlist you can audit

Orchestrator (gemini-4-argon) → ChEMBL Priming + Offline ML Ranker → Research Pool (gemini-3.8-flash) → 8 Python Gates → Human Sign-Off → JSONL Receipt

Held-Out Results (5 Seeds)

- Precision@20: 0.89 (89% top-20 hit rate)
- Precision@50: 0.61 (MCC@50: 0.71)
- AUPRC: 0.82 (vs 0.028 random baseline)

Hit prevalence: 3.3% (172 / 5,228 pairs).
Assay reproducibility capped at ~51% by replicate noise.

Calibrated Evidence Tiers

- Tier 1: Pair measured in another cancer
  Precision: 0.89 | 24.0x lift
- Tier 2: Both drugs known, pair untested
  Precision: 0.34 | 11.0x lift over 2.8% base rate
- Tier 3: Single-drug data only
  Precision: 0.06 | 3.0x lift
- Tier 4: Unseen chemical space
  Precision: 0.04 | REFUSED ('Test monotherapy first')

External Check (132 Merck pairs): +0.27 rank correlation (p=0.002). Honest negative result: direct NCI lookup beat ML (50% vs 40%), proving bounded transfer.

LIVE DEMO

The agent at work

Interactive 3D Synergy Landscape

Live interactive 3D dose-reachability viewer embedded on port 5173.

- Visualizes 5,228 measured pairs across 9 lung cancer cell lines
- Models dose-reachability surfaces over Bliss independence
- Replays empirical NCI-ALMANAC measurements (zero synthetic patient data)

Live command: ./demo.sh 1
Wi-Fi offline backup: ./demo.sh --replay

Live 4-Step Execution

1. 3D Landscape
Displays all 5,228 measured combinations.

2. Judge Prompt
'Which partner should a lab screen with gefitinib next?'

3. Multi-Agent Reasoning
PubChem resolves SMILES; ChEMBL primes EGFR targets; Gemini searches PubMed; 8 Python gates verify.

4. Human Scientist Sign-Off
System halts. Scientist approves batch; cryptographic JSONL receipt sealed.

Live demo runs on presenter laptop. Models never calculate numbers: every score comes from offline ML ranker.

THE BUSINESS CASE

Lower cost per real hit [Tentative]

Strategy	Pairs Tested	Screening Cost	Real Hits	Cost Per Real Hit
Test every pair (Full Screen)	5,228	$16.0M	172	$93,000 / hit
Random 50 (Intuition / Literature)	50	$153,000	~1.7	$93,000 / hit
LungPair Top 50 (Tier 1: Cross-cancer)	**50**	**$153,000**	**31**	**$4,900 / hit (18x lift)**
LungPair Top 20 (Tier 2: Brand-new pairs)	20	$61,200	~7	$8,700 / hit (11x lift)

Decision Analysis & NICE Cost-Effectiveness Threshold (£30,000/QALY)

- Technical vs Allocative Efficiency: Compresses cost per validated hit from $93k to $4.9k, dominating random screening in Quadrant II of the Cost-Effectiveness Plane.
- Downstream NICE Alignment: De-risking generic repurposed combinations lowers capitalized R&D attrition, ensuring therapies launch comfortably below NICE's £30k/QALY ceiling.

REGULATORY & SAFETY

Research tool now. Regulated tool later.

Now: Research Tool

Strictly Research Use Only (RUO).
Runs on public cell-line data.
Zero patient data ingested.
No clinical procurement drag.

Later: Medical Device (SaMD)

Clinical decision support from serial blood ctDNA.
Regulated as Software as a Medical Device (MHRA / FDA).
Requires prospective clinical validation.

8 Deterministic Python Gates (Plain Code, Not an LLM)

1. Pre-Model Intent Gate: Intercepts patient dosing queries before any LLM token is spent
2. Biologics Refusal: Deterministic refusal for antibodies (pembrolizumab) and unmapped drugs
3. Grounded Citations Gate: Any PMID/DOI not verified in search redirects is automatically stripped
4. Exact-Match Numbers Gate: Natural language decimals must match ranker output bit-for-bit
5. Calibrated Tier 4 Refusal: Unseen chemical space withheld ('Test monotherapy first')
6. Prompt Sanitization: Deterministic protection against prompt injection and system leakage
7. Human Sign-Off: Cannot order compounds or run assays without explicit scientist sign-off
8. Offline Replay Engine: Zero-latency execution (--replay) with 25 passing offline unit tests

ROADMAP · NOT BUILT

Next: re-rank before resistance sets in

Serial Blood ctDNA Monitoring Loop

1. Blood ctDNA Profiling: Serial liquid biopsy at baseline and set intervals.

2. Resistance Mutation Flagged: New mutations detected (e.g. KEAP1, STK11, EGFR T790M).

3. LungPair Re-Ranks Pairs: Ranks combinations prioritizing multi-cell line coverage.

4. Clinician / Researcher Sign-Off: Human review of candidate doublet hypotheses.

5. Lab Validation: Preclinical test results feed back into continuous learning loop.

Why Multi-Line Coverage?

18 pairs demonstrated synergy across ALL 9 tested lung cancer cell lines.

Tracking serial mutations lets us re-rank combinations before clinical resistance sets in, maximizing Expected Utility.

LungPair.

Fewer experiments, more real hits.

Team Backprop to the Future